BookStore Database Design

Database

Database Name: bookstore

Database Type: MongoDB Atlas

Collections Overview

The BookStore application contains the following collections:

1. users
2. books
3. categories
4. carts
5. wishlists
6. orders
7. reviews
8. addresses

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1. Users Collection

Stores customer and administrator information.

Field	Type	Required
_id	ObjectId	Yes
fullName	String	Yes
email	String (Unique)	Yes
password	String	Yes
phone	String	No
role	String (customer/admin)	Yes
profileImage	String	No
isActive	Boolean	Yes
createdAt	Date	Yes
updatedAt	Date	Yes

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2. Categories Collection

Stores book categories.

Field	Type
_id	ObjectId
name	String
description	String
createdAt	Date
updatedAt	Date

Examples:

- Fiction
- Programming
- History
- Science
- Business
- Self Help

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3. Books Collection

Stores all books.

Field	Type
_id	ObjectId
title	String
author	String
description	String
isbn	String
publisher	String
language	String
pages	Number
categoryId	ObjectId
price	Number
stock	Number

Field	Type
image	String
averageRating	Number
totalReviews	Number
featured	Boolean
createdAt	Date
updatedAt	Date

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4. Cart Collection

Stores customer cart items.

Field	Type
_id	ObjectId
userId	ObjectId
items	Array

Each item contains:

- bookId
- quantity
- price

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5. Wishlist Collection

Stores wishlist books.

Field	Type
_id	ObjectId
userId	ObjectId
books	Array
createdAt	Date

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6. Orders Collection

Stores customer orders.

Field	Type
_id	ObjectId
userId	ObjectId
orderNumber	String
items	Array
totalAmount	Number
shippingAddress	Object
paymentStatus	String
orderStatus	String
createdAt	Date

Order Status

- Pending
- Confirmed
- Shipped
- Delivered
- Cancelled

Payment Status

- Pending
- Paid
- Failed

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7. Reviews Collection

Stores customer reviews.

Field	Type
_id	ObjectId
userId	ObjectId
bookId	ObjectId
rating	Number

Field	Type
review	String
createdAt	Date

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8. Addresses Collection

Stores shipping addresses.

Field	Type
_id	ObjectId
userId	ObjectId
fullName	String
mobile	String
houseNumber	String
street	String
city	String
state	String
postalCode	String
country	String
isDefault	Boolean

Relationships

User

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Cart

↓

Order

↓

Review

Books

↓

Category

↓

Reviews

User

↓

Wishlist

↓

Books

Indexes

Users

- email

Books

- title
- author
- categoryId

Orders

- userId
- orderNumber

Reviews

- bookId
- userId

Estimated Collection Count

Collection	Expected Records
Users	5,000+
Books	10,000+
Categories	50+
Orders	50,000+
Reviews	100,000+
Wishlist	5,000+
Cart	5,000+

Database Flow

Customer

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Register

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Users Collection

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Browse Books

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Books Collection

↓

Add to Cart

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Cart Collection

↓

Checkout

↓

Orders Collection

↓

Write Review

↓

Reviews Collection

Future Collections

These can be added later without changing the existing design.

- coupons
- notifications
- banners
- publishers
- authors
- inventory
- paymentTransactions
- auditLogs
- activityLogs
- recommendations

Database Goals

- Normalized where appropriate while keeping MongoDB document efficiency.
- Easy to scale.
- Supports future features.
- Optimized for fast searches.
- Suitable for a real-world MERN application.